Safe, Gentle Seed Removal from Strawberries: simulation report

EN.601.497/697 Human-Centered Robotics, Fall 2026. Team: Kai Hua Liu, Zijun Xiang, Zixuan Zhu.

This report documents the simulation part of the project, step by step, with a screenshot for every step. Every image and number below is produced by the code in this repository (`scripts/make_report_assets.py`, `scripts/run_experiments.py`, `scripts/make_figures.py`, `scripts/make_report.py`). Nothing was run on a real robot.

Summary

A 7-joint arm carries a Wuji Hand 2. The hand holds tweezers and closes them with thumb and index finger. A wrist camera finds the seeds, a nominal planner removes them one by one, and a safety filter overrides commands that would leave the safe set. All results are over 8 random berries per condition, 8 target seeds each (64 seeds per condition).

Question from the proposal	Short answer from the simulation
1. Safe set: can a filter keep the arm and hand inside it?	Yes, in the nominal simulation. With the filter 0% of seeds had a violation and 100% were removed. A planner mis-tuned to press too deep and approach too fast violated the safe set on 100% of seeds; behind the filter the same planner violated it on 2% and removed 89% (without the filter: 67%).
2. Tool use: can the hand hold and squeeze the tweezers, and what can it sense?	Thumb and index close the jaws through contact. From finger encoders alone the jaw gap is known to 0.06 mm (rms). From fingertip force the squeeze on a 1 mm object is known to 0.014 N (rms).
3. Uncertainty: how conservative must the filter be?	With the well-tuned planner the margins made no difference: 0% violations at every setting. They matter when a mis-tuned planner makes the filter decide the depth. Violations then: 48% with no margins, 25% with 50% conformal bounds, 2% with 90% bounds, 0% with 99% bounds (100% of seeds removed). Three times the 99% bounds was too conservative: 0% removed.
4. Shared control: how to split authority?	Person picks the seed and steers sideways, robot controls depth and gripping, filter on: 100% removed, 0% with a violation, 10.0 s per seed. Person steering everything: 78% removed, 66% with a violation, 11.3 s per seed without the filter, 27% removed, 6% with a violation, 21.1 s per seed with it.
5. Sim-to-real: does the simulation-calibrated safe set transfer?	Only tested against a second, shifted simulation (softer fruit, noisier sensors), not on hardware. Not as it is: with the simulation calibration 19% of seeds had a violation (97% removed). Partial on-site calibration made the robot too cautious to grip (30% and 3% removed). With the full on-site calibration: 80% removed, 2% with a violation.

Main limits of this work are listed at the end. The most important ones: the fruit is not a finite-element body, the human operator is a model, and "sim-to-real" has not touched hardware.

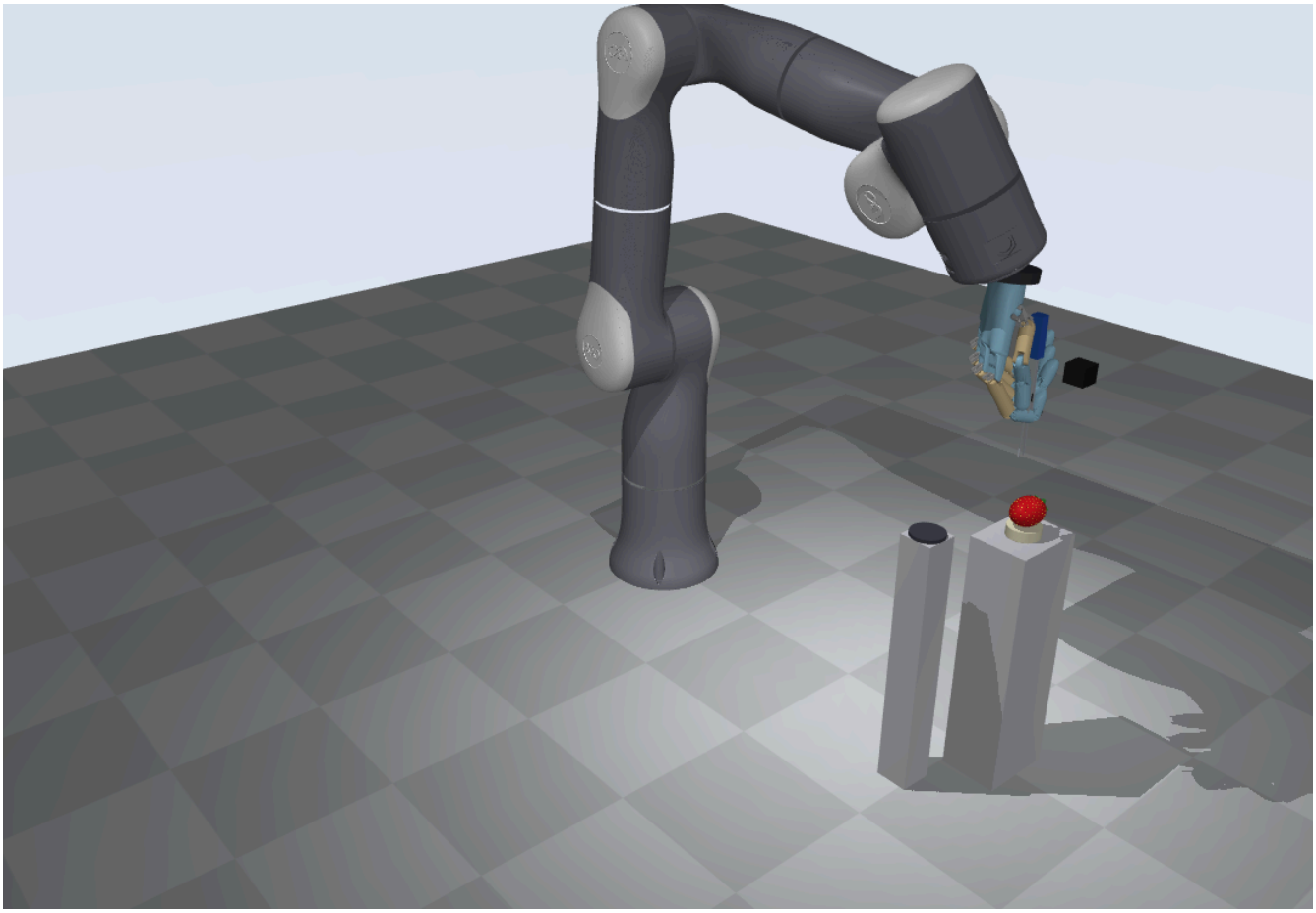
What from the proposal is implemented

Proposal item	Status	Where
MuJoCo scene: arm, Wuji Hand 2, tweezers, compliant berry, seed targets	done	`seedsafe/scene.py`, `seedsafe/env.py`
Force and indentation damage measure	done	`seedsafe/contact.py`
Scripted grasp that holds the tweezers and closes the jaws	done	`scripts/design_grasp.py`, `seedsafe/hand.py`
Nominal planner for arm and fingers	done (iLQR-based MPC)	`seedsafe/ilqr.py`, `seedsafe/planner.py`
Safety filter	done (CBF constraints in a QP, reachability-based approach speed)	`seedsafe/safety.py`
Seed localization with uncertainty bounds	done (wrist camera + conformal prediction)	`seedsafe/perception.py`, `seedsafe/conformal.py`
Tweezer pose and tip-force estimation from the finger joints	done	`seedsafe/hand.py`, `seedsafe/estimation.py`
Experiments against an unfiltered baseline	done	`scripts/run_experiments.py`
Human operator, shared autonomy	done with a simulated operator; a keyboard teleoperation script exists for real people	`seedsafe/operator.py`, `scripts/teleop_keyboard.py`
Calibrate the tweezer grasp away from the berry	done in simulation	`seedsafe/hand.py`
Port to the physical arm, proxy berry, real strawberries	not done (simulation only, as agreed)	

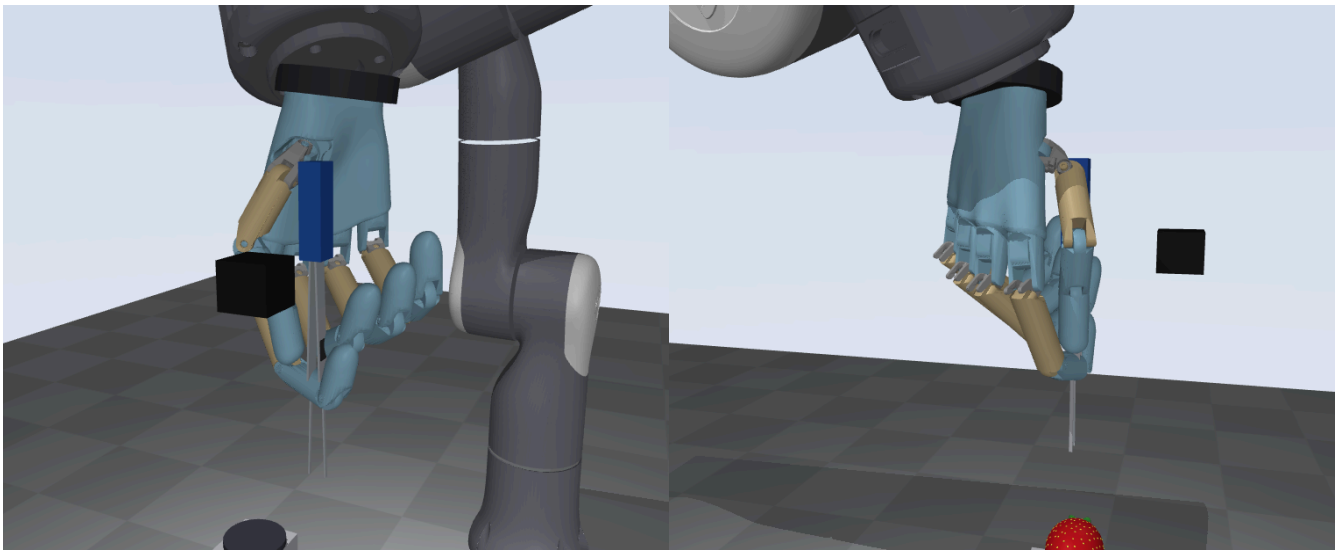
Abbreviations used below, with their full names: MPC (Model Predictive Control, 模型预测控制), iLQR (iterative Linear Quadratic Regulator, 迭代线性二次型调节器), CBF (Control Barrier Function, 控制屏障函数), QP (Quadratic Program, 二次规划), DoF (Degrees of Freedom, 自由度), MJCF (MuJoCo XML model Format, MuJoCo 模型文件格式), rms (root mean square, 均方根). A glossary of the technical terms (English / 中文) is at the end.

Step 1. The scene

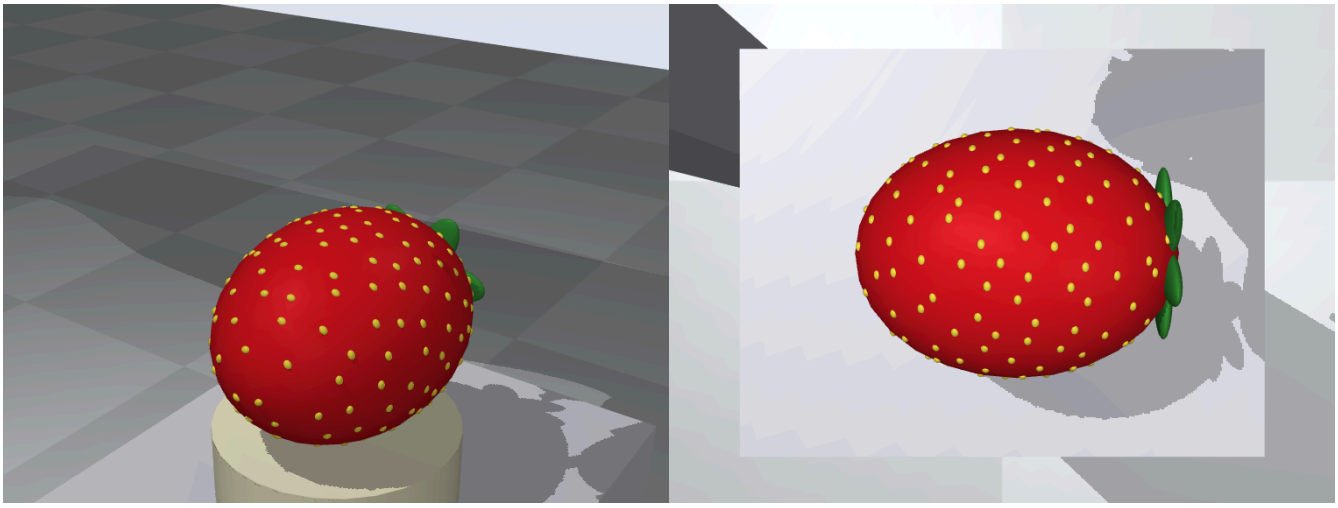
What was done. The official Wuji Hand model (20 DoF) is mounted on a Flexiv Rizon 4 arm model (7 DoF) through an angled adapter. The Rizon 4 stands in for the Flexiv Enlight because no public Enlight model exists. A pair of tweezers (two elastic arms joined at the base, 140 mm long) sits in the hand. A strawberry with about 200 seeds rests on a soft cradle; a tray takes the removed seeds. The model has 38 position coordinates and 27 motors.



The hand holding the tweezers, from two sides. Thumb and index finger pinch the two tweezer arms. The other three fingers wrap the printed sleeve (blue). The black box is the wrist camera.



Left: the berry. Right: what the wrist camera sees from its survey position.



Modelling choices that matter:

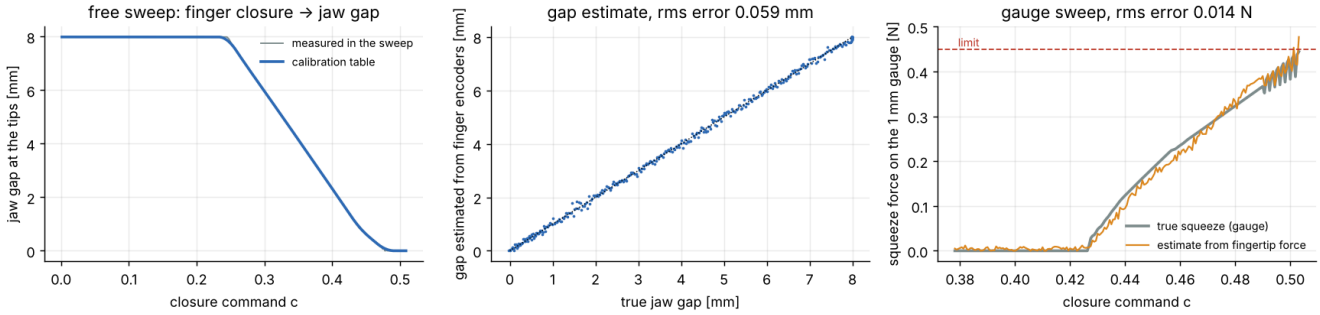
- **Fingers on the tweezers.** Thumb and index press the tweezer arms through simulated contacts. The pinch pose and a table "closure command c → finger joint targets" are designed once by `scripts/design_grasp.py`. The stock finger position gains are very soft, so the two pinching fingers use higher gains, and the thumb is made as stiff as the index along the pinch axis.
- **Sleeve in the hand.** The sleeve is held by a stiff seat that slips when the load exceeds the grip friction (6 N or 0.15 N m). Slip stays; the robot does not see it. This is the "tool must not slip" part of the safe set.
- **Arm.** Position-controlled joints with a task-space controller on top. The stock position gains of the arm model (about 1.5 Hz) are raised to about 6 Hz.

Step 2. Tweezer calibration (away from the berry)

What was done. This is the first hardware step of the proposal, done here in simulation. Two sweeps, run once per session:

1. *Free sweep.* Close the empty jaws slowly. Record the jaw gap and the thumb-index distance computed from the finger encoders. The jaws meet at closure c = 0.48.
2. *Gauge sweep.* Close on a rigid 1 mm ball and record the squeeze force.

After that the robot estimates the jaw gap from finger encoders (rms error 0.059 mm) and the squeeze force from the fingertip force sensors (rms error 0.014 N on the gauge). The sweep also showed that the two arms do not close evenly: the midpoint between the tips drifts sideways by up to 0.42 mm. That drift is stored and compensated. Without the compensation the tips landed next to the seeds.



Reproduce: `python scripts/make_report_assets.py calibration`

Step 3. Contact model, safe set and damage measure

What was done. MuJoCo simulates the arm, the hand, the tweezers and the finger contacts. The millimetre-scale contact at the two tweezer tips is computed by `seedsafe/contact.py`, because the safe set is defined on exactly these quantities:

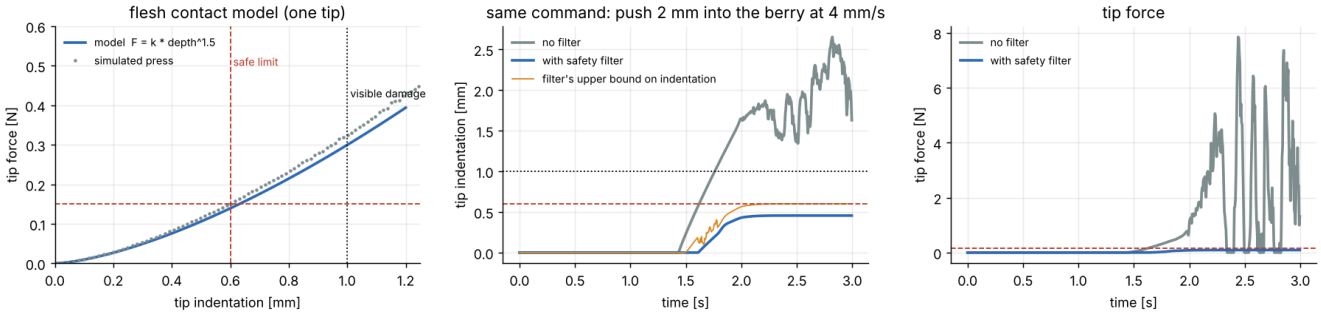
- *tip on flesh*: normal force `F = k * depth^1.5` (a Hertz-type law with damping) plus a shear spring that slides at the friction limit;
- *tip on seed*: stiff contact against a 1 mm sphere; the tips are flat pads 1.2 mm wide;
- *seed in its pit*: a spring that lets go at the pull-out force (about 0.08 N). The seed centre sits 0.35 mm below the surface, so the tips must press slightly into the flesh to grip it.

Safe set (per seed, judged on ground truth):

Quantity	Limit
tip indentation	0.6 mm
normal force per tip	0.15 N
shear force per tip	0.08 N
jaw squeeze on the seed	0.45 N
tool slip inside the hand	0.4 mm

Indentation beyond 1.0 mm or force beyond 0.3 N counts as *visible damage*.

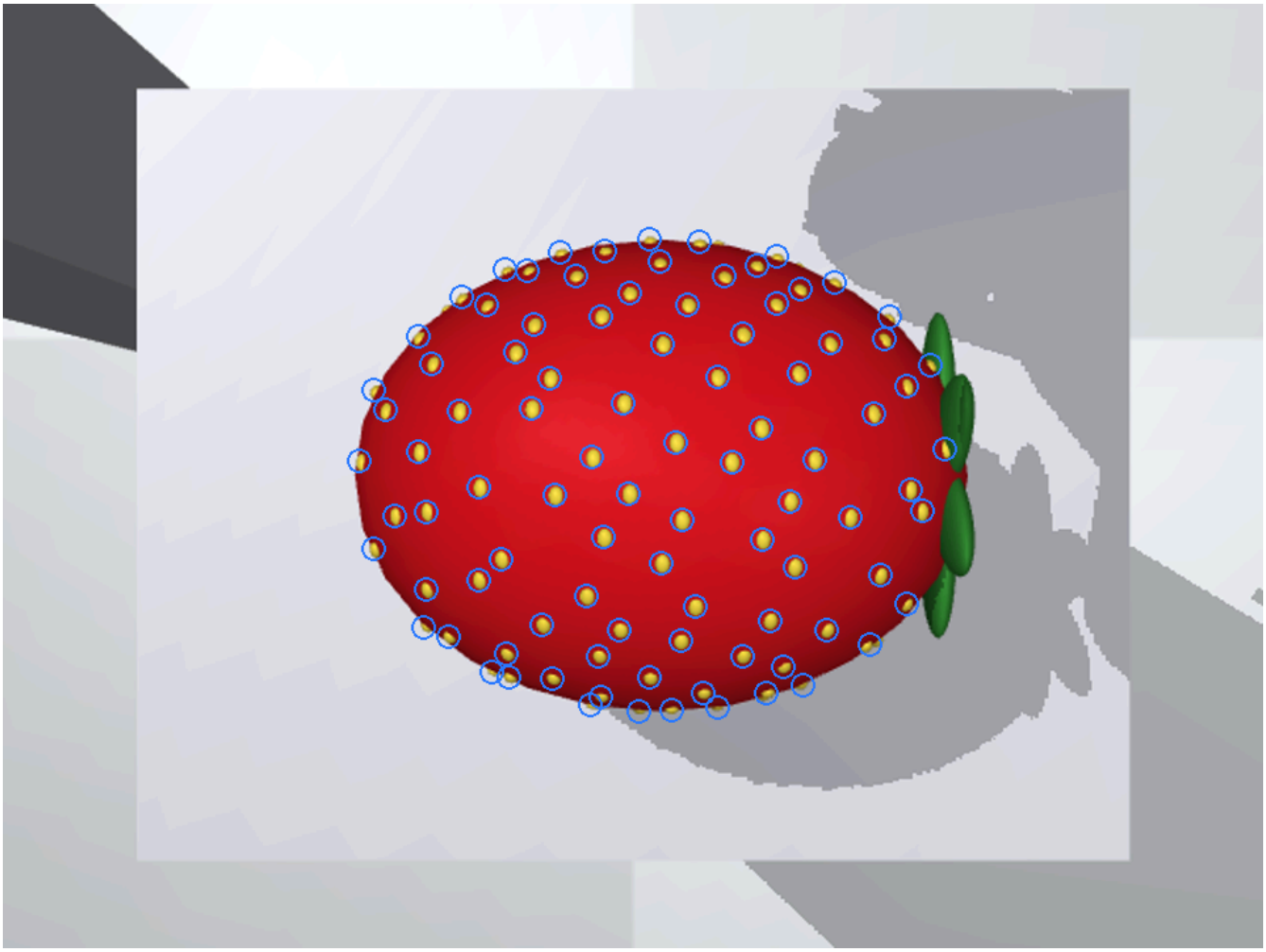
The figure shows the contact law (left) and what the same bad command does without and with the safety filter (middle, right): "push 2 mm into the berry at 4 mm/s". Without the filter the tips went 2.7 mm deep (7.9 N). With the filter they stopped at 0.46 mm (0.09 N).



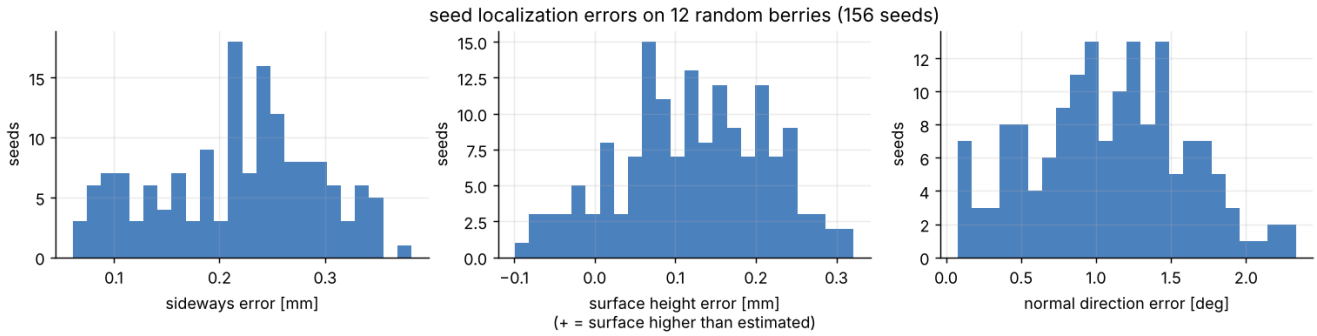
Reproduce: `python scripts/make_report_assets.py contact`

Step 4. Seed localization with the wrist camera

What was done. The camera moves about 10 cm above the berry and takes one colour and one depth image. Yellow blobs are seeds. For each seed the flesh around it gives the surface point and the surface normal. Modelled error sources: pixel size, depth noise, hand-eye calibration error, shading.



On 12 random berries (156 seeds): sideways error 0.22 mm on average (0.31 mm at the 90th percentile), surface height error +0.12 mm on average (0.24 mm at the 90th percentile), normal direction error 1.1 degrees. The height error has a bias: the camera places the surface a little too low.



Only the seeds inside a fixed patch on top of the berry are physically simulated and targeted (the proposal's "fixed patch of seeds").

Reproduce: `python scripts/make_report_assets.py perception`

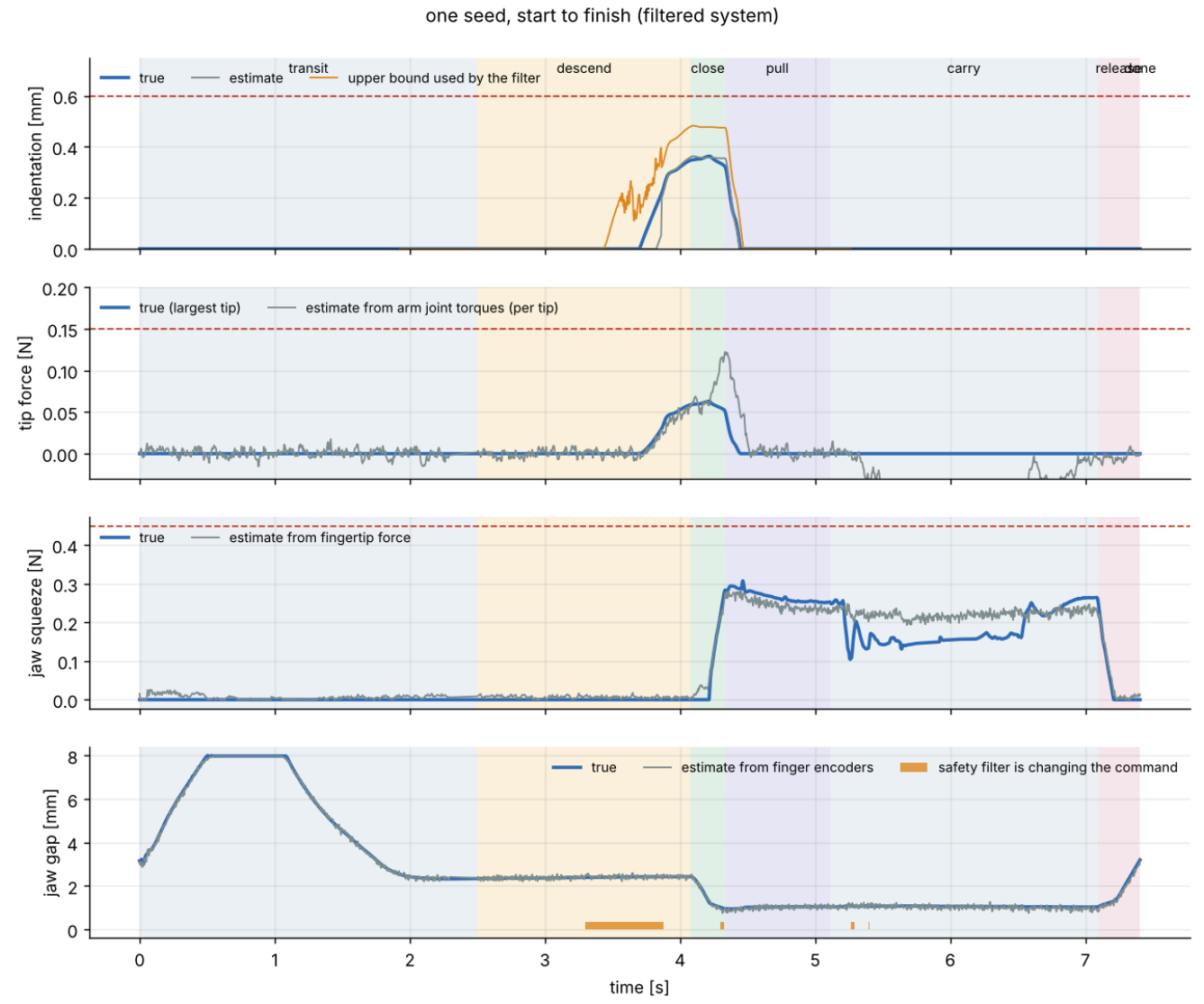
Step 5. Nominal planner and one seed from start to finish

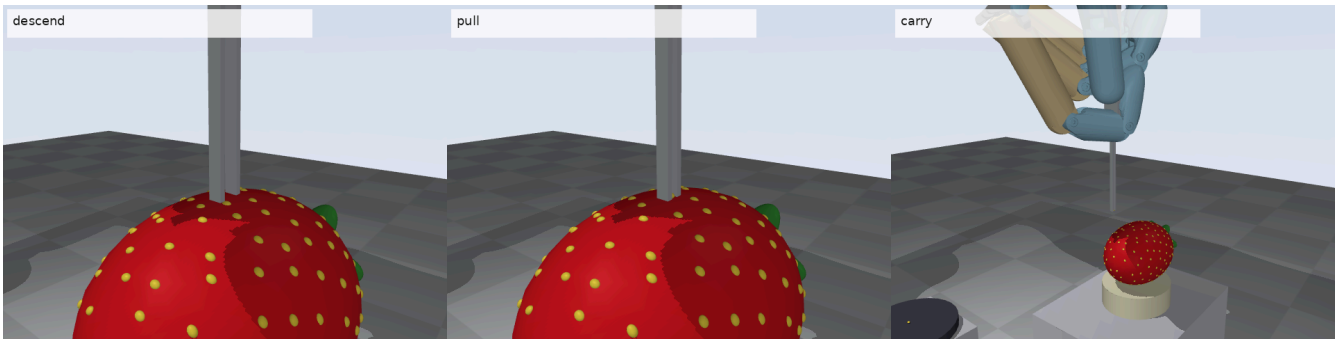
What was done. The planner is a phase machine: move above the seed, descend, close the jaws, pull, carry to the tray, release. Inside each phase the motion comes from MPC: an iLQR solver plans 0.5 s ahead on a small model (tip position and jaw closure), 50 times per second, with costs for reaching the goal, speed limits and staying clear of the berry between seeds.

The planner does not know the safe set. Two details make it gentle anyway:

- *Touch-referenced depth.* The camera's surface height is not accurate enough (Step 4). The planner descends slowly until the force estimate reports touch, and measures the gripping depth from there.
- *Force estimate.* Tip force is estimated from the arm's joint torque sensors and zeroed in free space before each descent.

The timeline below shows one seed with the filtered system: 7.4 s, peak indentation 0.36 mm, peak tip force 0.063 N, peak squeeze 0.31 N. Blue is ground truth, grey is what the robot estimates, orange is the upper bound the filter uses.





Reproduce: `python scripts/make_report_assets.py single`

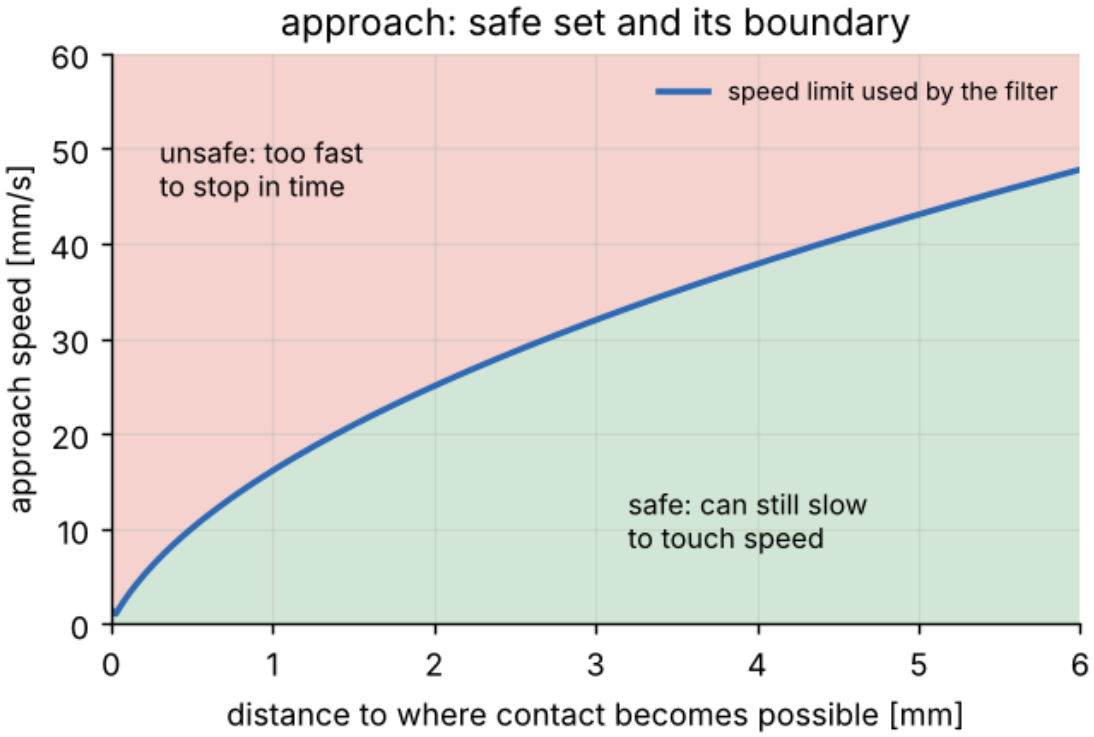
Step 6. Safety filter

What was done. The filter sits between the planner (or the human) and the robot. Every 5 ms it solves a small QP: stay as close as possible to the commanded tip velocity and jaw closing rate, subject to one CBF constraint per limit

of the safe set.

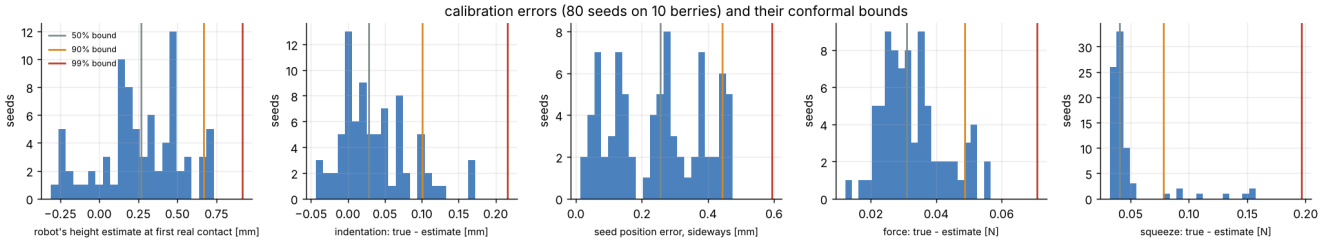
Constraint	Uses	Effect
indentation	upper bound on indentation (kinematics + touch, capped by what the measured force allows)	slows and stops the descent
normal force	force estimate + error bound, upper bound on flesh stiffness	slows, stops, or backs off
approach speed	distance to where contact becomes possible	tip arrives slowly enough to stop on touch
drag	"contact possible" flag	limits sideways motion while touching
grip stability	sideways force estimate vs. what the sleeve grip can hold	prevents tool slip
jaw squeeze	squeeze estimate + error bound	limits jaw closing

The approach-speed limit is the boundary of a reachable set: the largest speed from which the tip can still slow down to touch speed before contact becomes possible, given a reaction delay and a braking limit. The figure shows the set computed on a grid and the closed-form boundary the filter uses.



Step 7. Uncertainty bounds from conformal prediction

What was done. The filter needs error bounds for its estimates. They come from split conformal prediction: run calibration berries, record the errors against ground truth, and take the quantile that gives the wanted coverage. 80 seeds on 10 calibration berries were used. At 90% coverage the bounds are: surface height before touch 0.67 mm, indentation after touch 0.10 mm, sideways seed position 0.44 mm, tip force 0.049 N, squeeze 0.078 N.

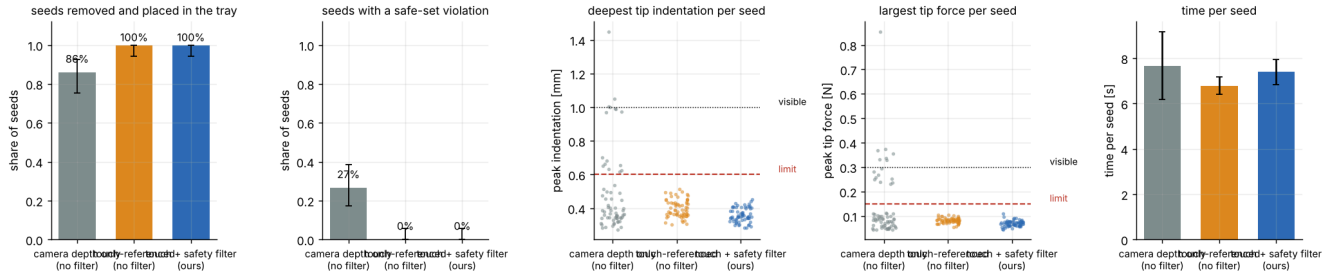


The sideways bound also sets how far the jaws are opened before descending, so that the seed is surely between the tips.

Step 8. Experiments

Each condition: 8 random berries (size, stiffness, seed layout, pull-out force, sensor biases and calibration errors all vary), 8 target seeds per berry. "Violation" means that at least one limit of the safe set was exceeded while working on that seed. Error bars are 95% Wilson intervals.

E1. Filtered system against unfiltered baselines



condition	seeds	removed	any violation	indent	force	shear	squeeze	slip	visible damage	time / seed [s]	attempts	filter intervention
camera depth only (no filter)	64	86%	27%	25%	25%	25%	14%	0%	13	7.7	1.34	0%
touch-referenced (no filter)	64	100%	0%	0%	0%	0%	0%	0%	0	6.8	1.00	0%
touch + safety filter (ours)	64	100%	0%	0%	0%	0%	0%	0%	0	7.4	1.02	1%

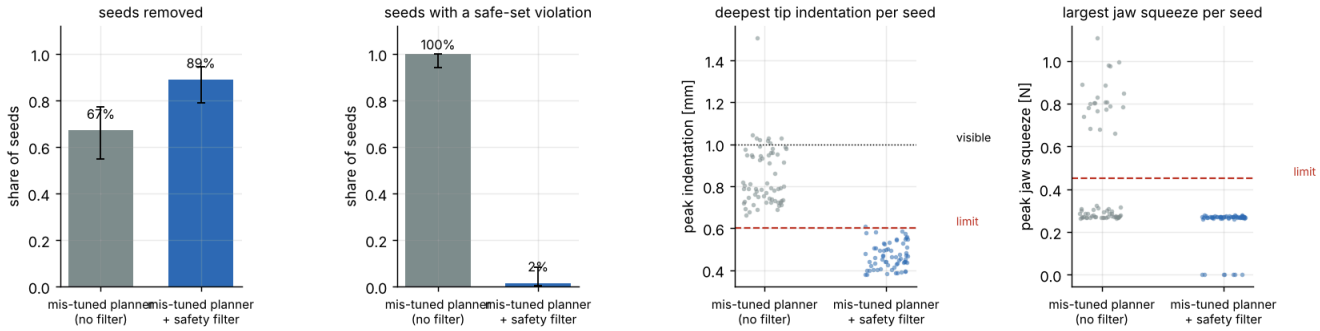
Findings:

- **Camera depth only, no filter:** 86% removed, 27% with a violation, 7.7 s per seed; 13 spots with visible damage. Relying on the camera for depth is not good enough.
- **Touch-referenced depth, no filter:** 100% removed, 0% with a violation, 6.8 s per seed. With a well-tuned planner and nominal conditions the planner alone stayed inside the safe set.
- **With the safety filter:** 100% removed, 0% with a violation, 7.4 s per seed. The filter changed 1% of the commanded motion and cost 7.4 s instead of 6.8 s per seed. No seeds were lost to the filter.

So in nominal conditions the filter is not what makes the system gentle; the touch-referenced planner already is. The filter matters when the command is wrong, which the next test shows.

E1b. A badly tuned planner, with and without the filter

The planner is told to go twice as deep (0.8 mm instead of 0.4 mm) and to approach four times faster (16 mm/s instead of 4 mm/s) than it should.



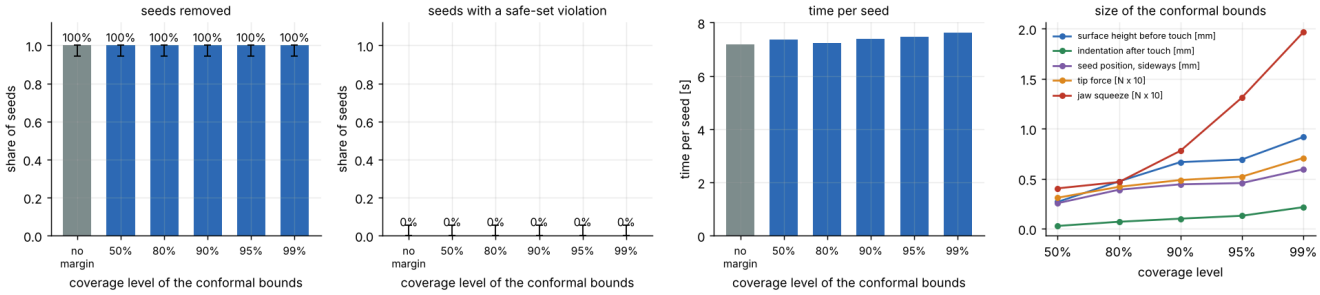
condition	seeds	removed	any violation	indent	force	shear	squeeze	slip	visible damage	time / seed [s]	attempts	filter intervention
mis-tuned planner (no filter)	64	67%	100%	100%	100%	100%	34%	0%	33	6.3	1.33	0%
mis-tuned planner + safety filter	64	89%	2%	2%	0%	0%	0%	0%	0	10.4	1.17	7%

Without the filter: 100% of seeds with a violation, deepest indentation 1.51 mm, 33 spots with visible damage, 67% of seeds removed (pressing too deep also makes the tips miss seeds). Behind the filter, with the 90% bounds: 2% of seeds with a violation (1 of 64), deepest indentation 0.61 mm against the limit of 0.6 mm, no visible damage, 89% of seeds removed. The filter changed 7% of the commanded motion and the time per seed went from 6.3 s to 10.4 s.

So the filter turned a planner that damages every seed site into one that almost never leaves the safe set. It is not a guarantee: one seed in this test went slightly past the limit.

E2. How conservative must the filter be?

The same system with uncertainty margins from conformal bounds at different coverage levels, and with no margins at all.

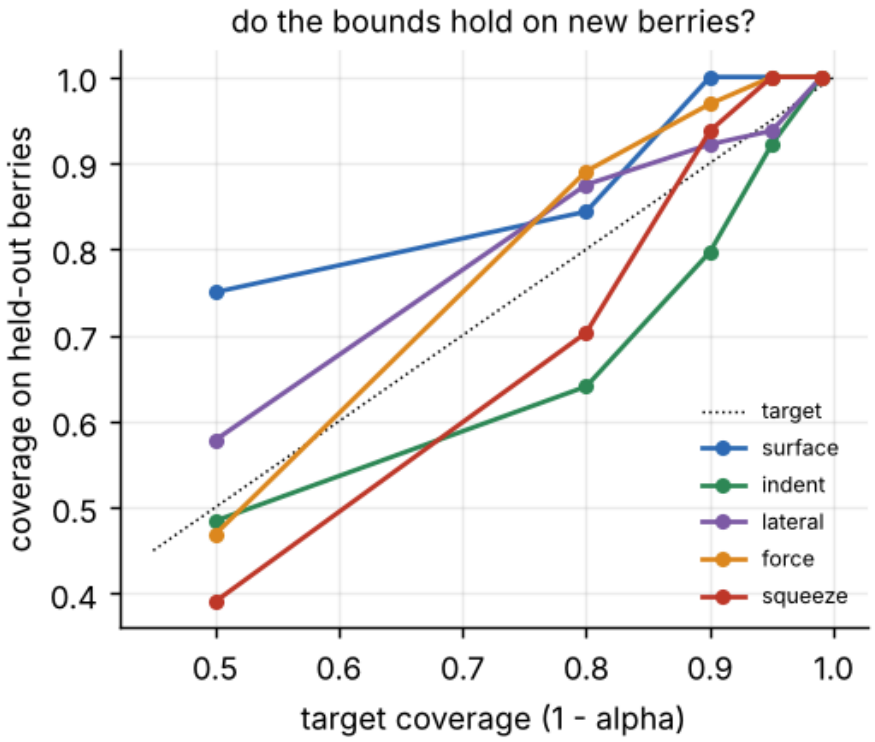


condition	seeds	removed	any violation	indent	force	shear	squeeze	slip	visible damage	time / seed [s]	attempts	filter intervention
no margin	64	100%	0%	0%	0%	0%	0%	0%	0	7.2	1.05	1%
50%	64	100%	0%	0%	0%	0%	0%	0%	0	7.4	1.08	1%
80%	64	100%	0%	0%	0%	0%	0%	0%	0	7.2	1.00	1%
90%	64	100%	0%	0%	0%	0%	0%	0%	0	7.4	1.02	1%
95%	64	100%	0%	0%	0%	0%	0%	0%	0	7.5	1.03	1%
99%	64	100%	0%	0%	0%	0%	0%	0%	0	7.6	1.02	2%

Bounds used:

coverage	surface [mm]	indentation [mm]	sideways [mm]	force [N]	squeeze [N]
50%	0.27	0.03	0.26	0.031	0.040
80%	0.47	0.07	0.39	0.042	0.047
90%	0.67	0.10	0.44	0.049	0.078
95%	0.69	0.13	0.46	0.052	0.131
99%	0.92	0.22	0.59	0.071	0.196

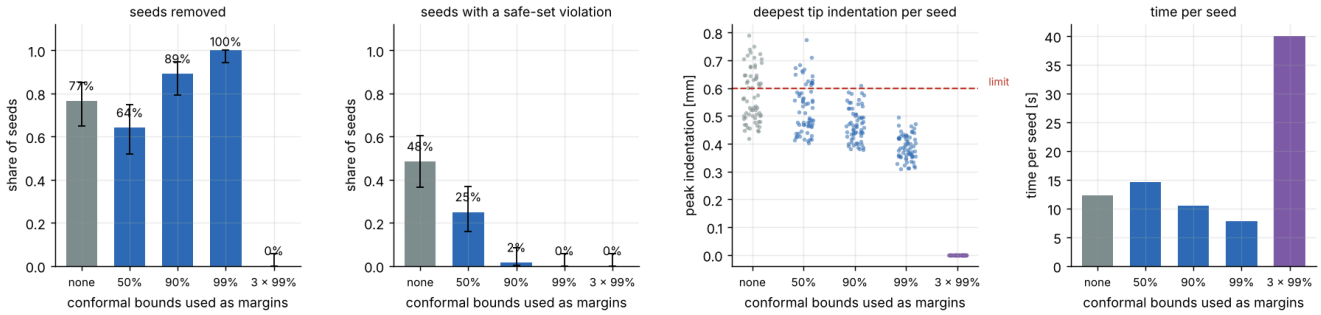
Coverage of the bounds on the test berries (they were calibrated on other berries):



Findings with the well-tuned planner:

- No margins: 100% removed, 0% with a violation, 7.2 s per seed. 50% bounds: 100% removed, 0% with a violation, 7.4 s per seed. 90% bounds: 100% removed, 0% with a violation, 7.4 s per seed. 99% bounds: 100% removed, 0% with a violation, 7.6 s per seed.
- Nothing changes except a little time. The reason is E1: the well-tuned planner is already gentle, so the filter rarely acts and its margins hardly matter.

To see what the margins do, the same sweep was repeated with the mis-tuned planner of E1b. Now the planner always asks for too much, and the filter decides how deep the tips go.



condition	seeds	removed	any violation	indent	force	shear	squeeze	slip	visible damage	time / seed [s]	attempts	filter intervention
no margins	64	77%	48%	44%	12%	3%	0%	0%	5	12.3	1.30	7%
50% bounds	64	64%	25%	25%	5%	5%	0%	0%	11	14.6	1.42	12%
90% bounds	64	89%	2%	2%	0%	0%	0%	0%	0	10.4	1.17	7%
99% bounds	64	100%	0%	0%	0%	0%	0%	0%	0	7.8	1.00	5%
3 x the 99% bounds	64	0%	0%	0%	0%	0%	0%	0%	0	40.0	1.00	0%

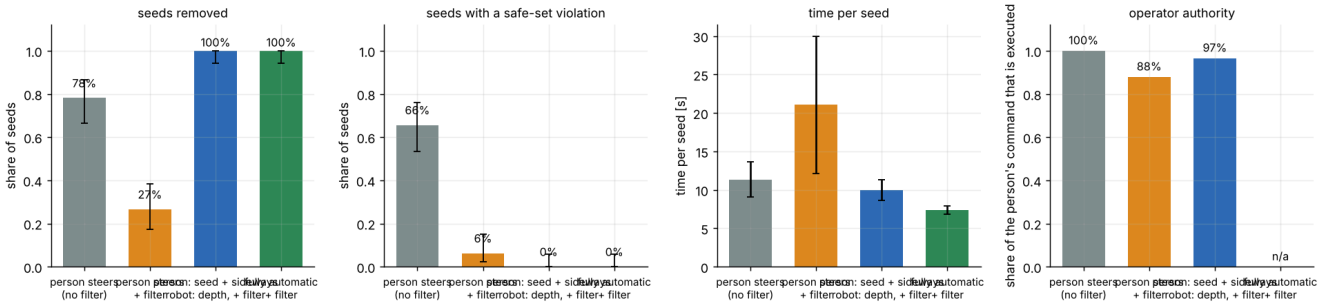
Findings when the filter decides the depth:

- **No margins:** 77% removed, 48% with a violation, 12.3 s per seed. The filter trusts its estimates exactly, and the estimates are a little off, so the tips often end up beyond the limit.
- **50% bounds:** 64% removed, 25% with a violation, 14.6 s per seed.
- **90% bounds:** 89% removed, 2% with a violation, 10.4 s per seed.
- **99% bounds:** 100% removed, 0% with a violation, 7.8 s per seed. This was the best setting in this test.
- **Three times the 99% bounds:** 0% removed, 0% with a violation, 40.0 s per seed. The force margin alone (0.21 N) is then larger than the force limit (0.15 N), so the filter never allows contact and every seed runs into the 40 s time limit.

Two cautions for reading the seed-removal numbers. First, the sideways bound also sets how far the jaws open before the descent. The 50% setting opens them less (0.26 mm bound instead of 0.59 mm at 99%), which by itself causes more misses. Second, each setting has 64 seeds on 8 berries, and failures cluster on single berries, so differences of 10 to 15 percentage points in removal between neighbouring settings are within the noise. The trend in violations (from 48% down to 0%) is the reliable part.

E3. Shared control: who commands what?

A simulated person steers the tip with a joystick model: reaction delay 0.2 s, tremor, motor noise, and depth judgement that is off by about 0.45 mm (standard deviation). Gripping is always automatic after the person presses "grip".



condition	seeds	removed	any violation	indent	force	shear	squeeze	slip	visible damage	time / seed [s]	attempts	operator authority
person steers (no filter)	64	78%	66%	64%	59%	55%	6%	0%	82	11.3	1.45	100%
person steers + filter	64	27%	6%	2%	6%	6%	0%	0%	3	21.1	1.77	88%
person: seed + sideways robot: depth, + filter	64	100%	0%	0%	0%	0%	0%	0%	0	10.0	1.08	97%
fully automatic + filter	64	100%	0%	0%	0%	0%	0%	0%	0	7.4	1.02	n/a

Findings:

- **Person steers everything, no filter:** 78% removed, 66% with a violation, 11.3 s per seed; 82 spots with visible damage. People judge depth poorly, so they press too deep.
- **Person steers everything, with filter:** 27% removed, 6% with a violation, 21.1 s per seed; 3 spots with visible damage. Violations drop from 66% to 6%, and 88% of the person's command is executed, but most seeds are missed. In the missed seeds the tips were deep enough (median 0.44 mm), yet no squeeze on the seed was recorded: the jaws did not close on it. The cause was not traced. The 4 seeds that still had a violation (tip force up to 0.34 N) were not traced either. A filter on top of full manual control was safer than no filter, but it was not a working system.
- **Person chooses the seed and steers sideways, robot controls depth, with filter:** 100% removed, 0% with a violation, 10.0 s per seed; 97% of the person's command is executed.
- **Fully automatic with filter:** 100% removed, 0% with a violation, 7.4 s per seed.

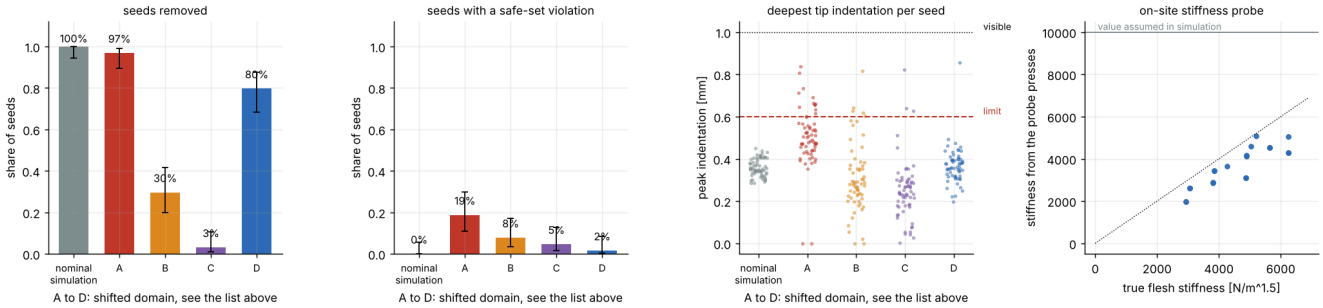
The split that worked: the person keeps the decisions and the sideways motion; the robot keeps depth, holding the tweezers and closing the jaws. The operator here is a model, so these numbers say what the control structure does, not how real people will perform.

E4. Transfer to a shifted domain ("sim-to-real" in simulation)

There is no hardware in this project, so transfer is tested against a second simulation that the robot was not calibrated in: flesh stiffness x0.42, seed friction 0.5 instead of 0.6, pull-out force x1.15, softer cradle, torque and fingertip sensor noise x1.5, depth noise x1.6, tool and hand-eye calibration errors x1.5 to x1.6.

- **A: simulation calibration only.** Model, thresholds and bounds from the nominal simulation.
- **B: + noise and stiffness probes.** Before the task the robot measures its force noise at rest (this sets the touch threshold) and presses gently on three seed-free spots to fit the flesh stiffness.
- **C: + surface re-referencing.** The same presses also show where the surface really is. That corrects the height bias of the camera.

- **D: + new conformal bounds.** C, plus bounds recomputed from 4 calibration berries in the shifted domain.



condition	seeds	removed	any violation	indent	force	shear	squeeze	slip	visible damage	time / seed [s]	attempts
nominal simulation (reference)	64	100%	0%	0%	0%	0%	0%	0%	0	7.4	1.02
A: simulation calibration only	64	97%	19%	19%	0%	0%	0%	0%	0	7.7	1.12
B: + noise and stiffness probes	64	30%	8%	8%	3%	3%	0%	0%	0	18.7	1.81
C: + surface re-referencing	64	3%	5%	5%	3%	3%	0%	0%	0	21.6	2.00
D: + new conformal bounds	64	80%	2%	2%	2%	2%	0%	0%	0	8.4	1.44

bound (90% coverage)	from the nominal simulation	recalibrated in the shifted domain
surface height before touch [mm]	0.666	0.163
indentation after touch [mm]	0.101	-0.049
seed position, sideways [mm]	0.444	0.530
tip force [N]	0.049	0.078
jaw squeeze [N]	0.078	0.136

Results: reference in the nominal simulation: 100% removed, 0% with a violation, 7.4 s per seed (64 seeds). A: 97% removed, 19% with a violation, 7.7 s per seed. B: 30% removed, 8% with a violation, 18.7 s per seed. C: 3% removed, 5% with a violation, 21.6 s per seed. D: 80% removed, 2% with a violation, 8.4 s per seed.

Reading the results:

- **A** removes almost everything (97%) but breaks the indentation limit on 19% of the seeds (deepest press 0.84 mm, limit 0.6 mm). The fruit is softer than the robot believes, so the same force means a deeper press and touch is felt later.
- **B** has fewer violations (8%) but removes only 30%. In the 45 missed seeds the tips went 0.25 mm deep (median); in the 19 gripped ones 0.39 mm. The tips stayed too shallow to grip, and after two attempts the planner gave up. The likely reason is the filter's indentation bound: on soft fruit the measured force says little about depth, so the filter depends on the surface-height bound, and that bound is wide.
- **C** does not help by itself (3% removed, 5% with a violation). The surface estimate is corrected, but the filter still uses the bounds from the simulation.
- **D** changes only the bounds compared with C. The surface-height bound, now measured after re-referencing, is 0.16 mm instead of 0.67 mm. The force and squeeze bounds grow (force 0.049 N to 0.078 N, squeeze 0.078 N to 0.136 N); the sensors are noisier in the shifted domain. Result: 80% removed, 2% with a violation (1 of 64 seeds, deepest press 0.86 mm), 8.4 s per seed.

The comparison of C and D shows that the bounds were what blocked the robot. Which bound mattered most was not separated. The sideways bound also grew, and it widens the jaw opening, which by itself reduces misses.

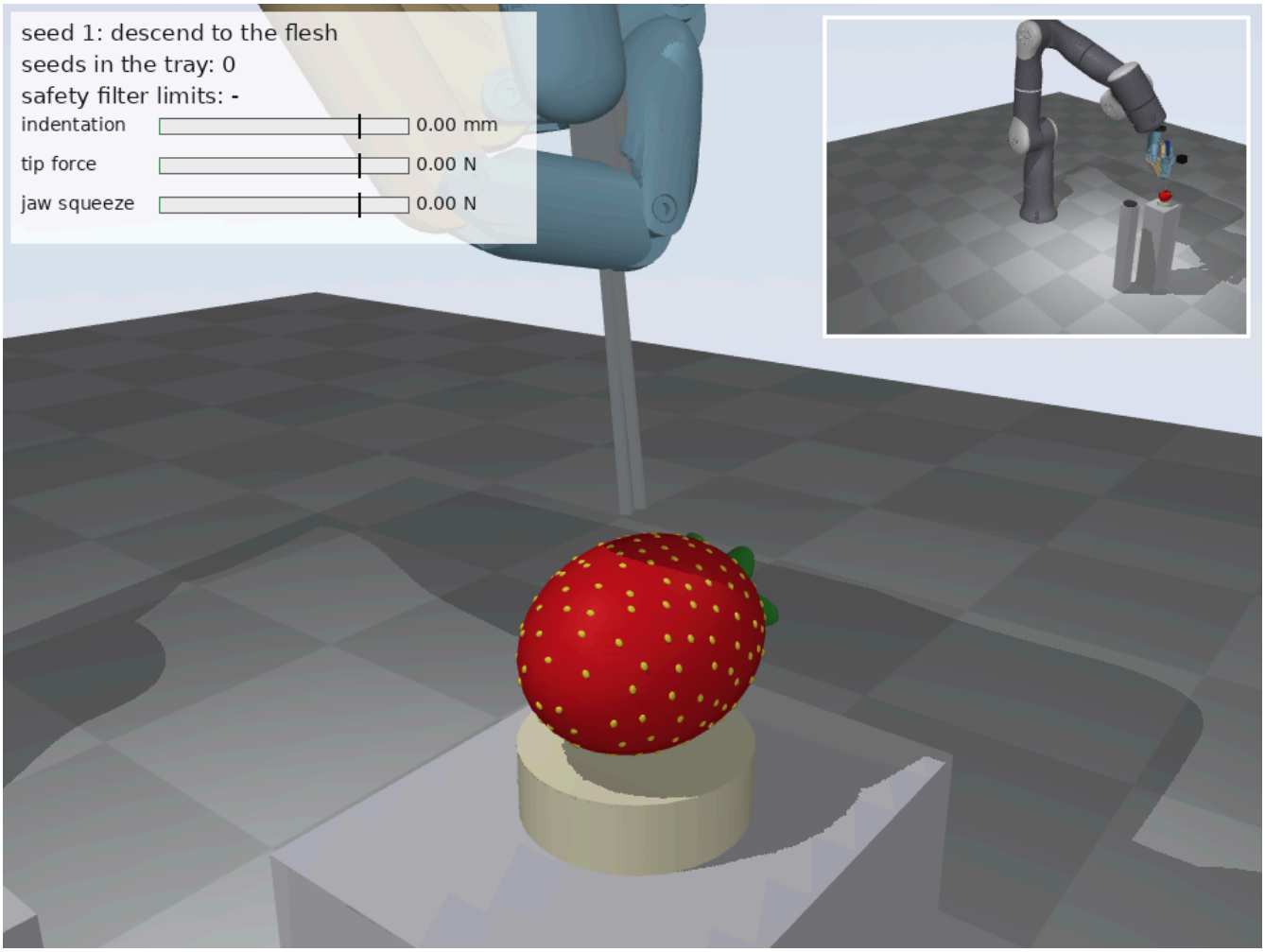
So the simulation-calibrated safe set did not transfer as it was. It transferred after an on-site calibration with four parts: force-noise check, stiffness probe, surface re-referencing and new conformal bounds. Even then the result (80% removed) is below the nominal simulation (100%).

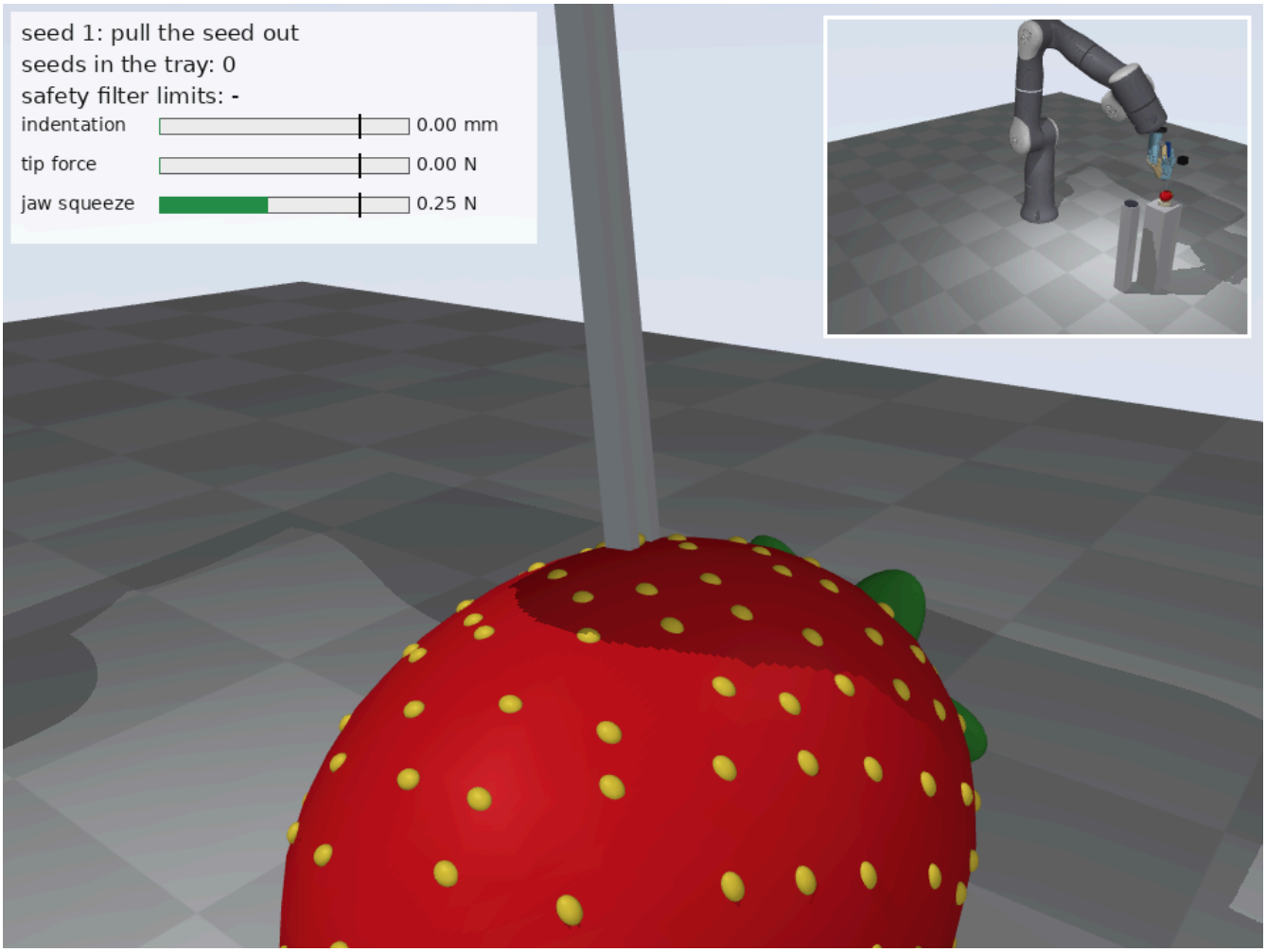
Cautions:

- **The probe itself touches the fruit.** Over 40 probed berries the probe presses went 0.49 mm deep (median) and at most 0.64 mm; on 10% of the berries they went past the 0.6 mm limit. The fitted stiffness was off by 19% (median).
- **The new bounds rest on little data.** They come from 4 berries (32 seeds; 22 for indentation and 12 for squeeze), collected while the robot was still missing most seeds. The indentation bound came out slightly negative (-0.05 mm), which means the robot's indentation estimate was above the truth in at least 90% of those cases.
- **The one violation in D was a deep one** (0.86 mm).
- **Simulation against simulation.** A real sim-to-real gap has effects that are not modelled here (see the limits below).

Step 9. Demo

`demo/seed_removal_demo.mp4` shows the filtered system removing six seeds from one berry, with live gauges for indentation, tip force and jaw squeeze against their limits. Key frames:





Reproduce: `python scripts/make_demo.py`

Step 10. Tests, and running on the laptop

`python -m pytest -q tests` runs 12 tests: the QP solver against brute force, conformal coverage, the berry geometry, iLQR, the reachable-set boundary, the tweezer calibration, "the filter stops a deep press", and a short filtered run that must remove seeds with no violation.

```
tests

$ python -m pytest -q tests
............                                                [100%]
12 passed in 22.97s
```

The same tests were also run on the team laptop in the Linux environment that the assistant can use there (Python 3.10, MuJoCo 3.17, no GPU rendering): 12 passed. The experiments and the video were produced on a cloud machine (Python 3.13, MuJoCo 4.0) and copied to the repository.

Limits and what is simplified

- **No hardware.** No real arm, hand, tweezers or fruit were used. The arm model is a Rizon 4, not an Enlight. The hand model is the public Wuji Hand description from the maker; it was not checked against the second-generation hardware.
- **Fruit model.** The berry is a rigid ellipsoid on a soft cradle. Deformation exists only as the contact law at the two tips. There is no finite-element flesh, no pits, no tearing. Stiffness, pull-out force and damage limits are plausible guesses, not measurements.

- **Sleeve grip.** The sleeve is held by a friction-limited seat, not by simulated finger contacts. Thumb and index contacts are simulated.
- **Sensing.** Tip force comes from joint torques with white noise and a slow bias; real torque sensing has friction and model errors that are not simulated. The noise level assumed here (about 0.01 N at the tip after filtering) is optimistic.
- **Human operator.** A model with delay, noise and depth error. No person took part. `scripts/teleop_keyboard.py` is ready for real trials but was not tested with a display.
- **Open questions in the results.** Three things were observed and not explained: why the jaws did not close on the seed when a person steers everything behind the filter (E3), which of the recalibrated bounds restored seed removal in the shifted domain (E4), and the few violations that the filter did not prevent (1 of 64 in E1b, 4 of 64 in E3, 1 of 64 in E4 D).
- **Statistics.** Each condition was run once. 64 seeds per condition. Rates near 0% or 100% have 95% intervals of about 0 to 6%.
- **Conformal bounds.** Calibration scores were collected with one filter setting; changing the margins changes the robot's behaviour a little, so the coverage guarantee is approximate. Seeds on the same berry share errors and are not fully independent.

Glossary (English / 中文)

Term	中文	Meaning here
achene	瘦果	the "seed" on the outside of a strawberry
end-effector	末端执行器	what the arm carries: here the hand with tweezers
degrees of freedom (DoF)	自由度	number of independent joints
forward kinematics	正运动学	computing where the tip is from the joint angles
Jacobian	雅可比矩阵	maps joint speeds to tip speed, and tip force to joint torques
indentation	压入深度	how far a tip is pressed into the flesh
safe set	安全集	all states in which every damage limit holds
safety filter	安全滤波器	module that changes a command as little as possible to stay in the safe set
control barrier function (CBF)	控制屏障函数	a function that is positive inside the safe set; the filter keeps it from decreasing too fast
quadratic program (QP)	二次规划	small optimization problem solved by the filter every 5 ms
reachability / reachable set	可达性 / 可达集	the states from which a failure can no longer be avoided (or can still be avoided)
model predictive control (MPC)	模型预测控制	plan a short time ahead, execute the first step, plan again
iterative LQR (iLQR)	迭代线性二次型调节器	the optimizer used inside the MPC
nominal planner	名义规划器	the planner that pursues the task and ignores the safe set
conformal prediction	保形预测	turns calibration errors into bounds with a coverage guarantee
coverage	覆盖率	share of new cases in which the bound holds
hand-eye calibration	手眼标定	where the camera sits relative to the robot
shared autonomy / shared control	共享自主 / 共享控制	person and robot command the motion together
teleoperation	遥操作	a person steers the robot directly
sim-to-real gap	仿真到现实差距	what works in simulation may behave differently on hardware
Hertz contact	赫兹接触	contact law in which force grows with depth to the power 1.5
tare	去皮 / 清零	setting the force reading to zero in free space
back-drivable	可反向驱动	a joint that outside forces can move, so its torque can be sensed

Last test run: `12 passed in 22.97s`